

Part 1: Theoretical Understanding (30%)

1. Short Answer Questions

Q1: Define *algorithmic bias* and provide two examples of how it manifests in AI systems.

- **Algorithmic bias** refers to situations in which artificial intelligence or machine-learning models generate unfair, uneven, or discriminatory outcomes in a consistent and predictable way.
- It learns from training data that may be incomplete, imbalanced, or influenced by historical inequalities.
- From the methods used to collect data or from the assumptions made by developers when building the system. Over time, these systems can reinforce harmful social, racial, gender-based, or economic disparities, especially when they create feedback loops that continue to favor certain groups while disadvantaging others.
- A common example is seen in hiring tools, where AI trained primarily on resumes from men ended up ranking male candidates higher, unintentionally replicating and amplifying gender discrimination. Another instance involves facial-recognition technologies, which have shown higher error rates for women and people with darker skin tones because their training datasets lacked sufficient diversity. Similarly, risk-assessment tools in the criminal justice system, such as COMPAS, have faced criticism for disproportionately rating Black individuals as high-risk, demonstrating how biased patterns in data can produce inequitable real-world consequences.

Q2: Explain the difference between *transparency* and *explainability* in AI. Why are both important?

Transparency in AI means openly sharing how the entire system works—its data sources, algorithms, and processes—making it accessible and accountable. Explainability focuses on clarifying why a specific AI decision was made, providing understandable reasons for individual outcomes. Both build trust and accountability in AI systems.

Q3: How does GDPR (General Data Protection Regulation) impact AI development in the EU?

The GDPR influences how AI is developed and deployed across the EU by imposing rigorous standards on the handling of personal information. It requires organizations to obtain clear consent before using individuals' personal or sensitive data for training AI systems. The regulation also emphasizes principles such as limiting data collection to what is necessary, maintaining transparency in how algorithms operate, and giving people the ability to understand and challenge automated decisions. While these rules can slow AI innovation by adding regulatory and compliance responsibilities, they also enhance public trust and promote responsible technology use. Recent updates within the EU aim to strike a better balance by slightly relaxing certain requirements to support AI advancement while still protecting user privacy. In essence, the GDPR plays a central role in steering AI development toward practices that prioritize ethics, openness, and strong privacy protections.

2. Ethical Principles Matching

Match the following principles to their definitions:

- A) **Justice** → Fair distribution of AI benefits and risks.
- B) **Non-maleficence** → Ensuring AI does not harm individuals or society.
- C) **Autonomy** → Respecting users' right to control their data and decisions.
- D) **Sustainability** → Designing AI to be environmentally friendly.

Part 2: Case Study Analysis (40%)

Case 1: Biased Hiring Tool

- **Scenario:** Amazon's AI recruiting tool penalized female candidates.
- **Tasks:**
 1. Identify the source of bias (e.g., training data, model design).
 2. Propose three fixes to make the tool fairer.
 3. Suggest metrics to evaluate fairness post-correction.

The bias in Amazon's hiring algorithm stemmed mainly from its training dataset, which consisted largely of resumes submitted by men—an outcome of longstanding gender disparities in the tech industry. Because the model learned from this skewed data, it began downgrading resumes that included indicators associated with women, such as references to “women's” organizations or women's colleges.

To make the system more equitable, several improvements could be applied:

- Train the model on a more balanced dataset that includes equal and diverse representation across genders.
- Remove gender-related terms and other demographic clues from resume inputs to reduce the risk of biased pattern learning.
- Apply fairness-focused techniques or bias-mitigation algorithms during model development to correct unequal patterns.

To evaluate whether fairness has improved, the following metrics can be used:

- **Statistical parity difference**, to compare selection rates among genders.
- **Equal opportunity difference**, measuring how true positive rates differ between groups.
- **Disparate impact ratio**, to gauge whether one group is being selected at a disproportionately higher or lower rate.

Case 2: Facial Recognition in Policing

- **Scenario:** A facial recognition system misidentifies minorities at higher rates.
- **Tasks:**
 1. Discuss ethical risks (e.g., wrongful arrests, privacy violations).
 2. Recommend policies for responsible deployment.

Misidentification of minority groups by facial recognition systems poses several ethical dangers. It can lead to innocent people being wrongfully detained or accused, eroding public confidence in law-enforcement institutions. It also raises privacy concerns, as minority communities often face heavier surveillance, creating unequal scrutiny. These issues reinforce existing racial inequities and threaten the fairness of the justice system.

Policies that support safer and more responsible use include:

- Restricting facial recognition to investigations of serious crimes under strong regulatory oversight.
- Conducting frequent accuracy and bias audits of the technology.
- Using training datasets that reflect diverse populations and ensuring openness about system performance.
- Requiring human verification before acting on AI-generated matches.
- Providing clear procedures for individuals to challenge incorrect identifications.
- Limiting intrusive monitoring in at-risk communities to protect civil rights and privacy.

These measures reduce potential harm and encourage a more ethical, transparent, and accountable application of facial recognition in policing.

Part 3: Practical Audit (25%)

Task: Audit a Dataset for Bias

1. Write a 300-word report summarizing findings and remediation steps.

Examination of the COMPAS dataset shows clear and persistent racial disparities in how defendants are assigned risk scores, with African-American individuals consistently receiving harsher assessments than their Caucasian counterparts. The fairness audit indicates that the algorithm is not operating equitably and that its current structure disproportionately burdens minority groups. A core metric, the disparate impact ratio, is roughly 0.65—far below the commonly accepted threshold of 0.8—demonstrating that African-American defendants face significantly higher odds of being labeled high risk. In fact, they are about 1.54 times more likely to receive high-risk classifications even when matched with Caucasian defendants who have similar backgrounds.

The error rate analysis further highlights these inequities. African-American defendants experience a false positive rate nearly twice that of Caucasian defendants, meaning they are incorrectly classified as high risk more often despite not reoffending. Conversely, Caucasian defendants show a somewhat higher false negative rate, reflecting a greater chance of being misclassified as low risk even when they do reoffend. A statistical parity difference of -0.18 reinforces the conclusion that risk scores are unevenly distributed along racial lines. Importantly, this imbalance remains evident even after adjusting for variables such as age and prior criminal history.

To address these issues, several remediation strategies are recommended. First, algorithmic techniques such as reweighing or adversarial debiasing can help correct imbalances in model outcomes. Second, a thorough review of input features is necessary to identify and remove any variables that may function as indirect indicators of race. Increasing transparency through explainable AI approaches can also improve accountability and allow human reviewers to understand and challenge risk classifications. Regular fairness monitoring should be established so that significant disparities trigger timely model updates. Finally, expanding fairness assessments to include multiple demographic dimensions—such as gender, age, and socioeconomic status—can provide a more comprehensive view of how the system impacts different groups.

Part 4: Ethical Reflection (5%)

- **Prompt:** Reflect on a personal project (past or future). How will you ensure it adheres to ethical AI principles?

Ethical AI Reflection: Brain Tumor Prediction System Promoting Responsible Practices in Medical AI Diagnostics

Data Responsibility and Patient Confidentiality:

Because medical imaging data is highly sensitive, I would apply strong anonymization procedures to remove all personal identifiers from the brain tumor dataset. The project would follow HIPAA data-protection requirements, ensuring that no confidential health information is revealed. All files would be securely encrypted, with access restricted only to qualified researchers who have undergone proper ethical training.

Fairness, Equity, and Bias Reduction:

To ensure the diagnostic model works accurately for all patients, I would perform comprehensive fairness assessments. This would include evaluating model performance across gender, age, and racial groups, checking for geographic imbalances in training data, integrating fairness constraints into the learning process, and routinely examining false-positive and false-negative rates for signs of unequal treatment.

Clinical Validation and Human Supervision:

The AI tool would function as a support system rather than a replacement for licensed medical professionals. Essential safeguards would involve requiring radiologists to confirm all AI-generated predictions, communicating the model's confidence levels and degree of uncertainty, establishing procedures for low-confidence cases, and ensuring that healthcare practitioners remain responsible for final diagnostic decisions.

Transparency and Model Explainability:

Given the critical nature of medical decision-making, the system would emphasize interpretability. This includes producing visual explanations highlighting the image regions that influenced predictions, presenting confidence and uncertainty scores, openly noting model limitations and potential failure scenarios, and providing thorough documentation for clinicians and technical stakeholders.

Patient Welfare, Safety, and Harm Prevention:

The system's core ethical principle would focus on maximizing patient well-being. This would involve strong validation to minimize false negatives that could delay treatment, carefully balancing sensitivity and specificity in a clinical context, conducting ongoing monitoring for performance decline, and establishing clear mechanisms for reporting errors and refining the model.

Regulatory Alignment and Professional Responsibility:

Development would comply with medical device regulations such as FDA guidelines for software used in healthcare and would involve collaboration with medical ethics boards. Clinical experts would participate throughout the process to ensure that the system meets established medical standards and supports high-quality patient care.

Bonus Task (Extra 10%)

- **Policy Proposal: Draft a 1-page guideline for *ethical AI use in healthcare*. Include:**
 - **Patient consent protocols.**
 - **Bias mitigation strategies.**
 - **Transparency requirements.**

Ethical Framework for Deploying Healthcare AI: Principles for Safe and Responsible Use

1. Patient Consent and Data Management

To support ethical deployment of AI in clinical environments, patients must be fully informed about how their data will be used and how AI contributes to their care. This involves obtaining explicit consent before applying AI tools in diagnosis and ensuring patients understand that AI serves as a support mechanism rather than a substitute for clinician expertise. Individuals should also have the option to decline AI involvement, and long-term data use should require periodic renewal of consent. Strong data governance is essential: all personally identifiable information must be thoroughly removed from training datasets, and sensitive health data should be encrypted during storage and transmission. Access rights should be strictly limited using role-based permissions, and only the minimum data necessary for clinical purposes should be collected.

2. Fairness, Bias Reduction, and Equity

Before an AI system is introduced into clinical workflows, it must undergo a comprehensive fairness audit. This includes evaluating the model's performance across demographic groups such as race, gender, age, and socioeconomic background, and ensuring the training data genuinely reflects the diversity of the patient population. Independent ethical oversight should review these evaluations. After deployment, continuous monitoring is required to detect performance drift and disparities. Any model that signals unfair patterns—through thresholds indicating unequal outcomes—should be paused and investigated. Remediation may involve retraining on more diverse datasets, conducting root-cause analyses, and notifying affected patients if harm is identified.

3. Transparency and Professional Accountability

Clear documentation is necessary to establish trust and facilitate responsible use. Technical reports should outline how the system works, what its limitations are, and how confident it is in its predictions. Known weaknesses and unusual edge cases must be documented. Clinicians should have access to interpretable outputs, including visual explanations for imaging models, confidence scores, and alternative diagnostic suggestions. A complete audit trail of AI recommendations and the human decisions that follow should be maintained. Communication materials should help patients and providers understand the AI's role, while institutions publicly disclose which systems are in use and provide regular updates to regulators.

4. Oversight, Governance, and Safe Integration

Strong governance is essential for responsible AI adoption. A multidisciplinary ethics committee should supervise system performance, while independent auditors periodically validate accuracy and safety. Clear accountability pathways must exist for any AI-related incidents, supported by structured response protocols. In clinical settings, AI should enhance—not replace—professional

judgment. Human review must be required for major findings, with procedures for escalation when clinicians disagree with AI assessments. Compliance with regulatory and privacy frameworks such as FDA SaMD rules, HIPAA, and GDPR is mandatory. Systems should also undergo routine security testing and meet certification standards.

- *This framework should be revisited at least annually to align with emerging technologies and evolving regulatory expectations. Above all, healthcare AI systems must prioritize patient protection, fairness, and high-quality clinical care rather than purely technical performance.*